

REVIEW

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Asymmetric effects of water resources on renewable energy in Morocco: evidence from the NARDL model

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Abstract

Understanding the relationship between water resources and renewable energy is increasingly important in countries facing growing pressure on water availability. This study examines the relationship between water abstraction and renewable electricity share in Morocco over the period 1990–2021. Gross domestic product (GDP), agricultural activity, and industrial activity are included as control variables. To account for potential asymmetries in the water–energy nexus, the analysis employs a Nonlinear Autoregressive Distributed Lag (NARDL) model and an asymmetric Granger causality approach. The bounds test results indicate the presence of a long-run cointegration relationship among the variables. However, the estimated long-run coefficients associated with positive and negative changes in water abstraction are statistically insignificant (−0.87 and 0.06, respectively), providing limited evidence of a direct long-run influence of water abstraction on renewable electricity share. In contrast, the short-run estimates reveal significant asymmetric dynamics. A negative change in water abstraction is associated with a substantial reduction in renewable electricity share (−5.28), whereas a positive change exerts a smaller positive effect (1.04). Agricultural activity displays a positive and statistically significant long-run relationship with renewable electricity share (5.28), while industrial activity does not exhibit a significant long-run effect. GDP shows a positive and statistically significant long-run effect (0.549, $p = 0.012$). The asymmetric Granger causality analysis further identifies a one-way predictive relationship running from negative changes in water abstraction to renewable electricity share. The findings suggest that the interaction between water abstraction and renewable electricity share is reflected primarily through asymmetric short-run adjustment mechanisms rather than through significant long-run effects. The stronger response associated with reductions in water abstraction indicates that changes in water resource conditions may influence renewable electricity dynamics over shorter horizons. These results provide valuable insights for policymakers seeking to balance water resource protection with the expansion of renewable electricity. They highlight the importance of integrating water resource considerations into renewable electricity planning, strengthening coordination between water and energy policies, and promoting a more balanced approach to sustainable resource management. In this respect, the study generates policy-relevant evidence that can support efforts to preserve water resources while advancing Morocco's renewable electricity transition in a sustainable and resilient manner.

Keywords Energy transition, Freshwater withdrawals, Renewable electricity, Morocco, NARDL, Water resource conditions

Introduction

The transition towards low-carbon energy systems has become a major priority within contemporary development agendas in response to increasing concerns regarding climate change, energy security, and

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environmental sustainability. International commitments aimed at reducing greenhouse gas emissions, coupled with the growing awareness of climate-related risks, have accelerated investments in renewable energy technologies, and stimulated the adoption of ambitious decarbonization strategies worldwide [1].

Renewable energy is increasingly recognized as a key pillar of sustainable development, contributing simultaneously to greenhouse gas mitigation, energy diversification, and long-term economic resilience [2]. Nevertheless, the success of energy transition pathways depends not only on technological innovation and financial investment but also on the sustainable management of critical natural resources, particularly water resources.

Water and energy systems are intrinsically interconnected through complex physical, economic, and environmental interactions. Energy production requires significant quantities of water for extraction, processing, cooling, and electricity generation activities, whereas water systems depend heavily on energy for abstraction, treatment, pumping, desalination, distribution, and wastewater management. This reciprocal relationship has given rise to the water–energy nexus framework, which has emerged as an increasingly important paradigm for understanding resource interdependencies and supporting integrated policy responses. As climate variability intensifies and pressure on freshwater resources continues to increase, ensuring the sustainable management of water and energy systems has become a strategic priority for policymakers and stakeholders worldwide.

Although renewable energy technologies are commonly regarded as environmentally sustainable alternatives to fossil fuel-based energy systems, they remain closely associated with water availability. Hydropower generation is inherently dependent on hydrological conditions, while concentrated solar power technologies, biomass production systems, desalination activities, and emerging green hydrogen industries require substantial water inputs throughout their operational processes. Consequently, increasing water scarcity and growing competition among economic sectors may affect renewable energy deployment trajectories and influence the long-term sustainability of energy transition strategies.

The importance of these interactions has stimulated a growing body of the literature devoted to the water–energy nexus. The previous studies have examined the implications of water scarcity for electricity generation, explored the contribution of renewable energy technologies to improving resource efficiency, and emphasized the importance of integrated planning approaches for enhancing sustainability outcomes. Existing research has also highlighted the necessity of coordinated policies capable of simultaneously addressing climate adaptation

challenges, water security concerns, and renewable energy development objectives.

Despite these advances, several important issues remain insufficiently explored. Existing investigations have predominantly focused on hydropower systems, water-use efficiency, and broader water–energy–food nexus interactions [3–6], while comparatively limited attention has been devoted to the relationship between freshwater abstraction and renewable electricity generation. Moreover, empirical evidence remains fragmented and often inconclusive regarding the extent to which water resources influence renewable energy development. While some studies suggest that favourable water conditions facilitate renewable energy deployment, others emphasize the dominant role of technological progress, institutional quality, and policy interventions. In addition, most empirical studies rely on conventional linear econometric specifications and implicitly assume that positive and negative changes in water-related variables exert identical effects on renewable energy systems. Such an assumption may oversimplify the complex dynamics characterizing resource-constrained economies and conceal important nonlinear adjustment mechanisms.

These limitations reveal an important scientific problem. Despite the rapid expansion of renewable energy systems worldwide, the extent to which changes in freshwater abstraction influence renewable electricity production remains insufficiently understood. Existing studies provide limited evidence regarding the impacts of freshwater withdrawals on renewable electricity dynamics and offer little insight into whether improvements and deteriorations in water conditions produce similar responses. Consequently, a fundamental question remains unanswered: do positive and negative variations in water abstraction generate symmetric or asymmetric effects on renewable electricity generation? Addressing this issue is particularly relevant for countries simultaneously pursuing ambitious energy transition agendas and experiencing increasing pressure on freshwater resources.

Morocco constitutes a particularly appropriate case study for investigating these questions. Over the past two decades, the country has established itself as one of the leading renewable energy actors in Africa through substantial investments in solar and wind infrastructures and the implementation of ambitious energy transition policies as shown in Figs. 1 and 2. National strategies seek to increase the contribution of renewable energy sources within the electricity mix while reducing dependence on imported fossil fuels. However, Morocco is also among the countries most exposed to water stress in the Mediterranean region, facing declining precipitation levels, recurrent drought episodes, rapid urbanization, demographic growth, and increasing demand from

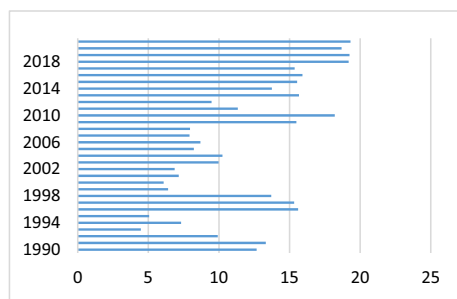


Fig. 1 Renewable electricity share in Morocco (proxy for RE). (Data source, WDI, 2025)

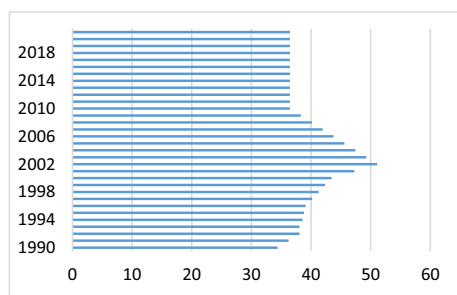


Fig. 2 Annual freshwater withdrawals in Morocco, total (% of internal resources). (Data source, WDI, 2025)

agricultural, industrial, and domestic sectors. Recent investments in desalination projects and green hydrogen initiatives have further strengthened the interdependencies between water resources and renewable energy development, thereby reinforcing the need for empirical evidence capable of supporting integrated resource management strategies.

Against this background, the present study seeks to answer three interrelated questions. First, does water abstraction significantly influence renewable electricity generation in Morocco? Second, do positive and negative changes in water abstraction exert asymmetric effects on renewable electricity share in the short run and the long run? Third, what are the implications of these potential nonlinearities for integrated water and energy management strategies in water-constrained economies? To address these questions, the study pursues three main objectives. First, it examines the long-run relationship between water abstraction and renewable electricity share in Morocco over the period 1990–2021. Second, it investigates whether positive and negative changes in water abstraction generate asymmetric responses in renewable electricity production. Third, it explores the direction of dynamic interactions between the variables through an asymmetric causality framework. Methodologically, the study

adopts the Nonlinear Autoregressive Distributed Lag (NARDL) model developed by Shin et al. [7], which enables explanatory variables to be decomposed into positive and negative partial sums and allows the identification of asymmetric short-run and long-run effects. Compared with conventional linear approaches, the NARDL framework provides a more comprehensive understanding of adjustment mechanisms and captures potential nonlinearities that may remain undetected in symmetric specifications.

This study contributes to the existing literature in four important ways. First, it enriches the relatively limited empirical literature investigating the relationship between freshwater abstraction and renewable electricity generation. Second, it provides novel evidence from Morocco, a country characterized by increasing water stress and ambitious renewable energy targets. Third, by employing a nonlinear econometric framework, the analysis identifies asymmetric responses that may remain overlooked in traditional linear models. Finally, the findings provide policy-relevant insights for designing integrated water and energy strategies capable of enhancing resource resilience and supporting sustainable energy transitions under conditions of increasing environmental uncertainty.

The remainder of the paper is organized as follows in Fig. 3. Section “Literature review and hypotheses development” reviews the relevant literature, identifies existing research gaps, and develops the research hypotheses. Section “Data and methodology” presents the data and methodological framework. Section “Empirical results and discussion” reports and discusses the empirical findings. Finally, Sect. “Conclusion and policy propositions” concludes by outlining the main policy implications, limitations, and avenues for future research.

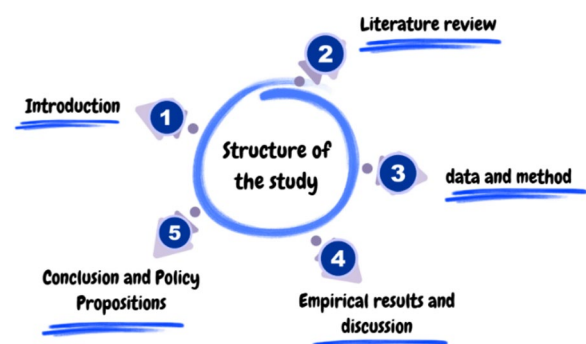


Fig. 3 Chart of article structure. Source: Performed by authors

Literature review and hypotheses development

This section reviews the existing literature on the interactions between water resources and renewable energy systems. It first discusses the theoretical foundations of the water–energy nexus, followed by an examination of empirical studies investigating the relationship between water availability and renewable energy development. Subsequently, the section identifies the main limitations and unresolved issues in the existing literature, thereby highlighting the research gap addressed by this study. Finally, based on the reviewed evidence, the research hypotheses are developed.

Water–energy interactions

Water and energy constitute two highly interconnected resources that are essential for economic development, environmental sustainability, and human well-being. Their mutual dependence has attracted increasing attention in recent years, particularly in the context of climate change, resource scarcity, and the global transition towards low-carbon economies. This interrelationship is commonly conceptualized through the water–energy nexus framework, which emphasizes the importance of integrated resource management and coordinated policy interventions [8, 9].

Water plays a critical role throughout the energy supply chain. Conventional energy systems require substantial quantities of water for fuel extraction, processing, refining activities, cooling operations, and electricity generation [10, 11]. Although renewable energy technologies are generally considered less resource-intensive than fossil fuel-based systems, several renewable pathways remain directly or indirectly dependent on water availability. Hydropower production relies fundamentally on hydrological conditions, while concentrated solar power technologies require water for cooling purposes [12–14]. Similarly, emerging sectors such as green hydrogen production and desalination involve significant water requirements, thereby reinforcing the interactions between water resources and energy development.

Conversely, water systems are highly dependent on energy availability. Energy inputs are necessary for water abstraction, treatment, pumping, transportation, distribution, wastewater collection, and desalination processes [15, 16]. In regions experiencing increasing water stress, energy-intensive technologies are becoming increasingly important for securing reliable water supplies. As a result, changes in energy systems may influence water accessibility, just as variations in water availability may affect energy production patterns.

These interdependencies are becoming increasingly relevant in countries pursuing ambitious renewable energy

transitions while simultaneously facing growing pressure on freshwater resources. Under such conditions, inadequate consideration of water constraints may hinder energy development, intensify competition among economic sectors, and compromise long-term sustainability objectives [17]. Therefore, understanding the interactions between water resources and renewable energy systems has become essential for designing integrated strategies capable of simultaneously enhancing water security, supporting climate adaptation, and promoting sustainable energy transitions. These interdependencies suggest that water availability should not merely be considered an environmental constraint but also as a strategic determinant of renewable energy transitions, especially in water-stressed economies [18].

Empirical evidence on water and renewable energy

Empirical investigations into the relationship between water resources and renewable energy development have attracted increasing scholarly attention in recent years [19–22]. This growing interest reflects mounting concerns regarding climate change, water scarcity, and the need to reconcile energy transition objectives with sustainable resource management. Nevertheless, existing evidence remains heterogeneous, revealing both converging and divergent findings across countries, sectors, and methodological approaches [23–26].

Several studies provide evidence supporting the existence of a positive relationship between water availability and renewable energy development. Favourable water conditions may facilitate renewable electricity production by improving the operation of water-dependent technologies, reducing resource constraints, and supporting investments associated with energy transitions. Using Egyptian data covering the period 1990–2022, Abdallah et al. [27] show that improved water resource conditions contribute positively to energy transition dynamics and stimulate renewable energy development. Similarly, Kousar et al. [28] report that greater water availability promotes renewable energy expansion while simultaneously improving environmental sustainability in South Asian economies. These findings suggest that water resources may act as an enabling factor for renewable energy deployment, particularly in countries where energy systems remain partially dependent on water availability.

Despite these findings, empirical evidence remains far from conclusive. Several studies argue that renewable energy development is primarily influenced by factors such as technological progress, institutional quality, investment capacity, and policy interventions, thereby reducing the relative importance of water-related determinants [29]. Furthermore, the

magnitude and direction of water–energy interactions appear to vary considerably across countries and regions depending on resource endowments, climatic conditions, and economic structures. [52] emphasize that renewable energy technologies cannot be considered entirely independent from water systems because many technologies require significant quantities of water during operation. Likewise, [50] underline that increasing water stress may represent a major challenge for future renewable energy deployment, particularly under conditions of growing climatic uncertainty and intensified competition over freshwater resources. These divergent findings indicate that the relationship between water resources and renewable energy systems remains context-specific and requires further empirical investigation. Beyond these mixed results, recent studies increasingly highlight the importance of nonlinear adjustment mechanisms within interconnected resource systems [30]. Most empirical analyses rely on conventional linear specifications and implicitly assume that positive and negative changes in water conditions exert symmetric effects on renewable energy systems. However, such assumptions may oversimplify the complex dynamics characterizing water-constrained economies. [51] for example, demonstrate that improvements in resource availability may generate stronger responses than equivalent deteriorations, suggesting that the direction of change constitutes an important determinant of system behaviour. Similarly, [58] argue that enhancing water efficiency within energy systems represents an important strategy for promoting sustainable energy transitions and mitigating environmental pressures. These findings suggest that positive and negative changes in freshwater abstraction may not necessarily affect renewable electricity production in the same manner, thereby supporting the relevance of nonlinear modelling approaches.

The existing literature provides valuable insights into the interactions between water resources and renewable energy systems. However, empirical evidence remains fragmented, particularly for developing economies undergoing ambitious energy transitions under increasing water stress. Moreover, the predominance of linear econometric frameworks limits our understanding of potential asymmetric responses associated with changes in water abstraction. This observation highlights the need for further research capable of simultaneously examining the existence, magnitude, and asymmetry of the relationship between water abstraction and renewable electricity generation.

Research gap

Despite the increasing attention devoted to the water–energy nexus, several important gaps remain in the existing literature. First, the previous empirical studies have primarily focused on hydropower systems, water-use efficiency, integrated resource management, and the environmental implications of energy production. Comparatively little attention has been paid to the relationship between freshwater abstraction and renewable electricity generation, despite the growing importance of water availability for sustaining energy transitions in water-constrained economies. Second, most existing investigations rely on conventional linear econometric approaches, implicitly assuming that increases and decreases in water-related indicators produce symmetric responses in renewable energy systems. Such an assumption may oversimplify the complex interactions between water resources and energy transitions, particularly in countries exposed to recurrent drought episodes, climatic variability, and increasing resource competition. Third, empirical evidence remains scarce for Morocco, even though the country has emerged as a regional leader in renewable energy development [31] while simultaneously facing increasing pressure on freshwater resources. Ambitious renewable energy targets, expanding desalination capacities, and the emergence of green hydrogen initiatives have considerably strengthened the interconnections between water and energy systems, thereby increasing the need for context-specific empirical evidence.

To address these shortcomings, this study investigates the relationship between water abstraction and renewable electricity share in Morocco using a nonlinear framework capable of capturing asymmetric responses in both the short run and the long run. By employing the NARDL approach, the analysis provides new insights into the role of water resource conditions in shaping renewable energy dynamics within a water stress economy undergoing an accelerated energy transition.

Hypotheses development

The empirical literature reviewed above suggests that water resources constitute an important component of sustainable energy systems. Nevertheless, the mechanisms through which water conditions influence renewable electricity development remain insufficiently understood. Existing evidence provides valuable insights into the interactions between water resources and energy systems, yet it does not establish a clear consensus regarding the magnitude, direction, and transmission channels of this relationship.

Several studies indicate that favourable water conditions can facilitate renewable energy deployment by reducing operational constraints, improving resource accessibility, and supporting investments associated with energy transitions. Abdallah et al. [27] demonstrate that improved water conditions contribute positively to energy transition dynamics in Egypt, while Kousar et al. [28] show that greater water availability promotes renewable energy expansion and environmental sustainability in South Asian economies. Similarly, [52] argue that renewable energy systems should not be examined independently from water resource conditions because several renewable technologies remain directly or indirectly dependent on water availability.

However, empirical findings remain fragmented. Previous investigations have predominantly focused on hydropower systems, water-use efficiency, and integrated resource management issues, whereas the relationship between freshwater abstraction and renewable electricity generation has received comparatively limited attention. In addition, most studies rely on aggregate indicators of water stress or water scarcity, providing little evidence regarding how changes in freshwater withdrawals may affect renewable electricity dynamics. This distinction is particularly important because water abstraction reflects evolving patterns of resource utilization and may capture multiple processes related to economic activity, technological progress, resource management practices, and environmental constraints.

From a theoretical perspective, variations in water abstraction may influence renewable electricity production through several channels. Increased water availability can facilitate the operation of water-dependent renewable technologies, improve resource accessibility, and support investments associated with energy transitions. Conversely, reductions in water withdrawals may signal increasing resource constraints capable of affecting electricity generation patterns, altering investment decisions, and reshaping long-term energy planning strategies. These considerations suggest that a statistically significant relationship between water abstraction and renewable electricity share is likely to exist. Accordingly, the following hypothesis is proposed:

H1 Water abstraction significantly influences renewable electricity share in Morocco.

While H1 examines the existence of a relationship between water abstraction and renewable electricity share, the possibility that this relationship may vary according to the direction of changes in water conditions remains largely unexplored. Existing studies have

predominantly relied on linear modelling frameworks, implicitly assuming that positive and negative variations in water-related indicators generate equivalent responses in renewable energy systems. However, such an assumption may oversimplify the complex dynamics characterizing water-constrained economies. In practice, improvements in water availability may not necessarily produce effects of the same magnitude as deteriorating water conditions, particularly in contexts characterized by recurrent drought episodes, increasing climate uncertainty, and growing competition over freshwater resources. This issue motivates the formulation of the second hypothesis concerning the potential asymmetric effects of water abstraction on renewable electricity generation.

Positive variations in water abstraction may reflect improved resource conditions, increased economic activity, or expanding sectoral demand, whereas negative changes may be associated with drought episodes, conservation policies, efficiency gains, or structural transformations in resource management. Consequently, renewable electricity systems may react differently depending on the direction and magnitude of changes in water conditions.

Recent methodological developments advocate the use of nonlinear frameworks to capture such heterogeneous responses. In particular, the Nonlinear Autoregressive Distributed Lag (NARDL) model proposed by Shin et al. [7] allows the decomposition of explanatory variables into positive and negative partial sums and provides estimates of asymmetric short-run and long-run effects. This methodology has gained increasing attention in environmental and energy economics, as well as in other applied economic domains such as trade forecasting [32], because it enables the identification of adjustment mechanisms that may remain undetected within conventional linear specifications.

Despite these methodological advances, empirical applications examining asymmetric relationships between freshwater abstraction and renewable electricity generation remain scarce. This gap is especially evident in countries characterized by increasing pressure on freshwater resources and ongoing renewable energy transitions. Investigating potential asymmetries, therefore, represents an important contribution to the existing literature and may provide valuable insights for designing integrated water and energy management strategies. Accordingly, the second hypothesis is formulated as follows:

H2 Positive and negative changes in water abstraction exert asymmetric effects on renewable electricity share in Morocco.

Related evidence on asymmetric environmental and energy responses using NARDL specifications includes Khaliq and Mamkhezri [33], who examine economic complexity and economic freedom in South Asia, and Khaliq et al. [34], who document asymmetric effects of electricity generation from renewable and fossil-fuel sources in Pakistan. Complementary studies on renewable energy policy and heterogeneous transition outcomes include Khaliq and Mamkhezri [35], on renewable portfolio standards and land-use governance across the US regions, and Khaliq et al. [36], on international clean energy R&D funding in Asian economies. Further NARDL-based evidence on asymmetric renewable energy and environmental responses is provided by Rahaman et al. [37] for South and East Asian economies, Oktay et al. [38] for China, Tlili et al. [39] for Saudi Arabia, and Miah et al. [40], who document asymmetric effects of rainfall on rice production in Bangladesh.

Data and methodology

Variables description and theoretical foundations

This study examines the relationship between water abstraction dynamics and renewable electricity share in Morocco. The analysis covers the period from 1990 to 2021 and is primarily based on data from the World Bank's World Development Indicators. Renewable electricity share (RE) is measured as the share of electricity generated from renewable sources in total electricity generation. This indicator is widely used in the literature as a proxy for renewable energy (e.g. Walaa et al. 2025). However, it should be noted that this measure captures only the electricity dimension of renewable energy and does not fully reflect broader adoption across all energy uses. Water abstraction (WA) represents total freshwater withdrawals from surface and groundwater sources and is used as an observable indicator of water use and resource conditions. Importantly, water abstraction is not interpreted as a direct measure of water scarcity. Changes in freshwater withdrawals may reflect multiple factors, including policy interventions, efficiency improvements, and structural changes in water demand. Within the NARDL framework, water abstraction is decomposed into positive and negative partial sums in

order to capture potential asymmetries. Positive changes in water abstraction (WA^+) represent increases in freshwater withdrawals, which may be associated with higher agricultural activity, industrial expansion, population growth, or rising economic demand. Negative changes in water abstraction (WA^-) represent reductions in freshwater withdrawals, which may reflect drought conditions, water-use restrictions, efficiency improvements, or structural adjustments in water-consuming sectors. Therefore, these positive and negative components should be interpreted as changes in water withdrawal patterns rather than direct measures of water abundance or water scarcity. To account for other factors that may influence renewable electricity share, the study includes several control variables. Gross domestic product (GDP), measured in constant 2015 US dollars, captures the overall level of economic activity. Agricultural value added (AGR) and industrial value added (IND), both expressed as shares of GDP, are included to reflect sector-specific dynamics that may influence energy demand and the energy transition. While GDP includes sectoral value-added components by construction, the simultaneous inclusion of GDP, agricultural value added, and industrial value added serves different analytical purposes. GDP captures overall macroeconomic conditions, whereas AGR and IND reflect structural and sector-specific dynamics that may affect renewable electricity output differently. This specification allows for a more nuanced understanding of the role of economic structure in the energy transition. Table 1 presents the variables used, along with their abbreviations, definitions, data sources, and roles.

Freshwater withdrawals are used as the main explanatory variable in this study, following previous research on water resource management and sustainability. As discussed earlier, water abstraction is treated as an observable indicator of water use and resource conditions rather than as a direct measure of water scarcity. The decomposition of water abstraction into positive and negative partial sums allows the analysis to capture potential asymmetric responses of renewable electricity share to changes in water withdrawal patterns. In addition to water, other factors can influence renewable

Table 1 Description of the variables. Source: Performed by authors

Variables	Abbr	Definition	Source	Role
Renewable energy	RE	Renewable electricity share (% of total electricity output)	WDI	Dependent
Water abstraction	WA	Annual freshwater withdrawals, total (% of internal resources)	WDI	Independent
Gross domestic product	GDP	GDP (constant 2015 US\$)	WDI	Control
Agriculture value added to GDP	AGR	Agriculture, forestry, and fishing, value added (% of GDP)	WDI	Control
Industrial value added to GDP	IND	Industry, including construction, value added (% of GDP)	WDI	Control

energy. First, gross domestic product plays a complex role in the energy transition. Some studies show that higher GDP can support sustainability and energy transitions, particularly in developing countries facing poverty and inequality [55, 56]. Economic growth provides resources for renewable energy investment, innovation, and long-term job creation. However, rapid growth can sometimes slow down the transition. Immediate economic priorities may conflict with the structural changes needed for clean energy. Still, this effect is usually temporary, because long-term sustainable energy investment ultimately boosts growth and innovation [49, 54]. Second, industrial growth also has mixed effects. Industry consumes large amounts of energy, so its expansion can provide an excellent opportunity to encourage renewable energy development through technological improvements and sustainable energy systems [59]. On the other hand, if industrial growth is not supported by energy investments and innovation, it may cause economic and energy challenges [53]. Third, agricultural growth matters for renewable energy. Agriculture consumes energy and water, and their development can either support or limit clean energy adoption. Studies show that sustainable agricultural practices and energy efficient farming can enhance renewable energy use, while resource intensive farming may increase pressure on water and energy systems [57, 60]. GDP, agricultural activity, and industrial activity are included as control variables to isolate the relationship between water abstraction and renewable electricity share. Their inclusion helps account for broader economic and sectoral influences that may affect the energy transition process.

Estimation method

To examine the effect of water resources on renewable electricity, this study specifies an empirical model linking water abstraction to renewable electricity share. In Eq. (1), we can present the relationship between the variables:

$$\log RE_t = \alpha_0 + \alpha_1 \log WA_t^+ + \alpha_2 \log WA_t^- + \alpha_3 \log GDP_t + \alpha_4 \log AGR_t + \alpha_5 \log IND_t + \varepsilon_t \quad (1)$$

In Eq. (1), RE is measured as the share of electricity generated from renewable sources. WA represents total freshwater withdrawals and captures water abstraction. To account for other factors that may influence RE, the study includes several control variables. GDP denotes gross domestic product, reflecting GDP (constant 2015 US\$), AGR represents agricultural value added, while IND refers to industrial value added as a share of GDP. The term α_0

is the intercept, and ε_t is the error term. The coefficients α_1 – α_5 measure the effect of each explanatory variable on renewable electricity. Variables are expressed in natural logarithmic form to improve comparability and allow for elasticity interpretation.

A comprehensive unit root analysis is first conducted to determine the order of integration of the variables used in the study. This step is essential to ensure that none of the series is integrated of order two I(2), which would invalidate the application of the NARDL model. For these reasons, the Augmented Dickey–Fuller [41, 42] and Phillips–Perron [43] unit root tests are employed. In both tests, the null hypothesis assumes the presence of a unit root, indicating non-stationarity, while the alternative hypothesis implies stationarity. To ensure the mixed order of integration, we applied unit root test with structural breaks too. Given the mixed order of integration of the variables, the study adopts the Nonlinear Autoregressive Distributed Lag (NARDL) approach proposed by Shin et al. [7]. This methodology is particularly appropriate when variables are integrated of order I(0) and I(1), and when the researcher aims to capture potential asymmetric effects in both the short run and the long run, asymmetric effects are formally tested using Wald tests after estimating the NARDL model. To allow for asymmetry, water abstraction is decomposed into its positive and negative partial sum components. This decomposition in Eqs. (3) and (4) makes it possible to distinguish between the effects of increases and decreases in water abstraction on renewable electricity. The partial sum processes are presented as follows:

$$\Delta \log(WA_t) = \Delta \log(WA_t)^+ + \Delta \log(WA_t)^- \quad (2)$$

where

$$\log(WA_t)^+ = \sum_{j=1}^t \max(\Delta \log(WA_j), 0) \quad (3)$$

$$\log(WA_t)^- = \sum_{j=1}^t \min(\Delta \log(WA_j), 0) \quad (4)$$

Here $\log(WA_t)^+$ captures cumulative positive changes in water abstraction, while $\log(WA_t)^-$ captures cumulative negative changes. This decomposition enables the identification of asymmetric responses of renewable electricity share to water-related shocks.

Based on this decomposition, the NARDL model in Eq. (5) is specified in its unrestricted error correction form as follows:

$$\begin{aligned} \Delta \log RE_t = & \alpha + \sum_{i=1}^p \phi_i \Delta \log RE_{t-i} + \sum_{i=0}^q (\varphi_i^+ \Delta \log WA_{t-i}^+ + \varphi_i^- \Delta \log WA_{t-i}^-) \\ & + \sum_{i=0}^r \gamma_i \Delta \log GDP_{t-i} + \sum_{i=0}^s \delta_i \Delta \log AGR_{t-i} + \sum_{i=0}^u \psi_i \Delta \log IND_{t-i} \\ & + \lambda_1 \log RE_{t-1} + \lambda_2 \log WA_{t-1}^+ + \lambda_3 \log WA_{t-1}^- + \lambda_4 \log GDP_{t-1} \\ & + \lambda_5 \log AGR_{t-1} + \lambda_6 \log IND_{t-1} + \varepsilon_t \end{aligned} \quad (5)$$

where Δ denotes the first difference operator, and ε_t is the error term. The coefficients associated with differenced variables capture short-run dynamics, while the coefficients on lagged level variables provide information about the long-run relationship. The optimal lag length for each variable is selected using the Akaike Information Criterion (AIC) and the Schwarz Bayesian Criterion (SBC) as shown in Fig. 8.

To examine the existence of a long-run cointegration relationship among the variables, the bounds testing approach is applied. The null hypothesis of no cointegration is formulated as follows:

$$H_0 : \lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = \lambda_5 = \lambda_6 = 0$$

Against the alternative hypothesis:

$$H_1 : \text{at least one } \lambda_i \neq 0$$

Rejection of the null hypothesis indicates the presence of a long-run equilibrium relationship between renewable energy, water abstraction, and the control variables. Once cointegration is confirmed, the long-run asymmetric coefficients are derived from the estimated NARDL model as follows:

$$\begin{aligned} \beta_{WA}^+ &= -\frac{\lambda_2}{\lambda_1}, \quad \beta_{WA}^- = -\frac{\lambda_3}{\lambda_1} \\ \beta_{GDP} &= -\frac{\lambda_4}{\lambda_1}, \quad \beta_{AGR} = -\frac{\lambda_5}{\lambda_1}, \quad \beta_{IND} = -\frac{\lambda_6}{\lambda_1} \end{aligned}$$

These coefficients represent the long-run asymmetric elasticities of renewable electricity with respect to positive and negative changes in water abstraction, as well as GDP, agricultural, and industrial activity.

The corresponding short-run dynamics are captured through the error correction representation:

$$\begin{aligned} \Delta \log RE_t = & \sum_{i=0}^q \varphi_i^+ \Delta \log WA_{t-i}^+ + \sum_{i=0}^q \varphi_i^- \Delta \log WA_{t-i}^- \\ & + \sum_{i=0}^r \gamma_i \Delta \log GDP_{t-i} + \sum_{i=0}^s \delta_i \Delta \log AGR_{t-i} \\ & + \sum_{i=0}^u \psi_i \Delta \log IND_{t-i} + \rho ECM_{t-1} + \varepsilon_t \end{aligned} \quad (6)$$

where ECM_{t-1} denotes the error correction term derived from the long-run relationship. The coefficient ρ is expected to be negative and statistically significant, reflecting the speed at which deviations from the long-run equilibrium are corrected.

$$\begin{aligned} ECM_{t-1} = & \log RE_{t-1} - \beta_{WA}^+ \log WA_{t-1}^+ - \beta_{WA}^- \log WA_{t-1}^- \\ & - \beta_{GDP} \log GDP_{t-1} - \beta_{AGR} \log AGR_{t-1} - \beta_{IND} \log IND_{t-1} \end{aligned} \quad (7)$$

To formally assess the presence of asymmetry, Wald tests are conducted for both long-run and short-run coefficients. Specifically, the null hypothesis of long-run symmetry is tested as follows:

$$H_0 : \beta_{WA}^+ = \beta_{WA}^-$$

Against the alternative hypothesis:

$$H_1 : \beta_{WA}^+ \neq \beta_{WA}^-$$

where β_{WA}^+ and β_{WA}^- denote the long-run coefficients derived from the estimated NARDL model. Short-run symmetry is assessed by testing the equality of the cumulative short-run coefficients:

$$H_0 : \sum \varphi_i^+ = \sum \varphi_i^-$$

Against the alternative hypothesis:

$$H_1 : \sum \varphi_i^+ \neq \sum \varphi_i^-$$

Rejection of these null hypotheses provides evidence of asymmetric short-run and/or long-run effects of water abstraction on renewable electricity.

In addition to estimating the asymmetric effects using NARDL, we apply asymmetric Granger causality test to examine the direction of influence between water abstraction and renewable electricity. This method extends the conventional Granger causality by allowing positive and negative shocks in the explanatory variable to exert different causal effects on the dependent variable. In line with this approach, water abstraction is decomposed into its positive and negative components, enabling the identification of asymmetric causal relationships.

The null hypothesis of the asymmetric Granger causality test states that positive or negative changes in water abstraction do not Granger-cause changes in renewable electricity, while the alternative hypothesis suggests the existence of asymmetric causality [44]. By examining whether past positive or negative shocks in one variable systematically precede movements in another, the test provides deeper insights into the direction and nature of asymmetric causal links. The causality analysis is conducted only after confirming the existence of a long-run cointegration relationship among the variables, in order to avoid spurious inference [7, 45]. The combined use of the NARDL model and asymmetric Granger causality analysis is particularly appropriate for understanding both the magnitude and the direction of asymmetric effects between water abstraction and renewable electricity. Finally, several diagnostic and stability tests are performed to ensure the robustness of the estimated model. Serial correlation is examined using the Breusch–Godfrey LM test, with the null hypothesis of no serial correlation, while heteroskedasticity is assessed using the ARCH test, where the null hypothesis assumes constant variance of the error terms [46]. Model specification is evaluated using the Ramsey RESET test. Parameter stability is assessed using the cumulative sum (CUSUM) and cumulative sum of squares (CUSUMSQ) tests proposed by Brown et al. [47]. If the test statistics remain within the 5% critical bounds, the null hypothesis of parameter stability cannot be rejected, indicating that the model is structurally stable over time. In the end, Fig. 4 resumes our processing of this study.

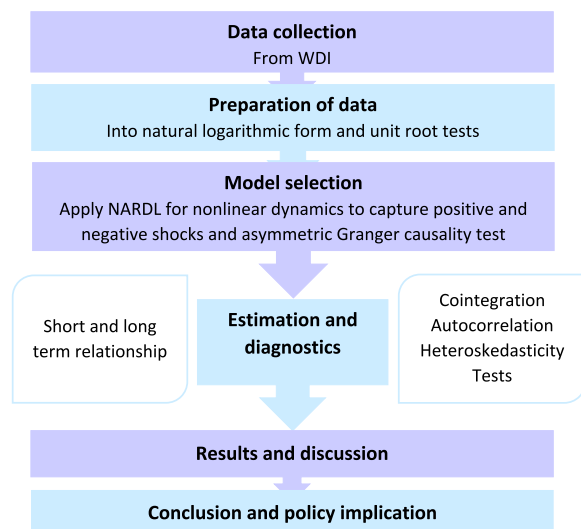


Fig. 4 Chart of methodology. Source: Performed by authors

Table 2 Descriptive statistics. Source: Authors' computation using EViews 12

Variable	Mean	Minimum	Maximum	Std. Dev	Observations
RE	12.00	4.47	19.32	4.61	32
WA	39.77	34.40	51.08	4.38	32
GDP	78.4	41.6	124.0	27.9	32
AGR	12.54	9.88	18.44	2.13	32
IND	24.74	22.26	27.60	1.46	32

Empirical results and discussion

Descriptive statistics

As a first step, descriptive statistics are presented to provide an overall picture of the variables used in the analysis. Table 2 reports the descriptive statistics for renewable electricity, water abstraction, economic growth, agricultural activity, and industrial activity over the period 1990–2021. Renewable electricity share averages 12%, indicating a moderate but gradually increasing role of renewable energy in Morocco's energy mix. Water abstraction remains relatively high, with an average of 39.8%, reflecting intensive pressure on water resources. Economic growth shows a clear upward trend, as GDP increases from 41.6 to 124.0 billion USD over the study period. Agricultural activity appears more volatile than industrial activity, while the industrial sector displays relatively stable values, suggesting a continued reliance on conventional energy sources.

Unit root tests

After descriptive statistics, this study first ensures the correct order of integration of the variables before estimating the model. Since the NARDL framework does not yield valid or reliable results when variables are

Table 3 Unit root test results (ADF and PP). Source: Authors' computation using EViews 12

Variable	At level		At 1st difference		Orders of integration
	ADF	PP	ADF	PP	
logRE	−2.07 ^C 0.25	−1.94 ^C 0.31	−4.78 ^C 0.00***	−8.22 ^C 0.00***	I(1)
logWA	−0.35 ^N 0.54	−0.01 ^N 0.67	−2.69 ^N 0.00***	−2.76 ^N 0.00***	I(1)
logGDP	−2.70 ^C 0.08*	0.90 ^C 0.99	−8.29 ^C 0.00***	−7.96 ^C 0.00***	I(1)
logAGR	−2.27 ^C 0.18	−2.97 ^C 0.04**	−11.56 ^C 0.00***	−14.89 ^C 0.00***	I(1)
logIND	−2.99 ^C 0.04**	−2.93 ^C 0.05**	−5.79 ^C 0.00***	−8.99 ^C 0.00***	I(0)

***, **, and * denote 1%, 5%, and 10% significance levels, respectively. C: with intercept. N: with no intercept and no trend

integrated of order two $I(2)$, it is necessary to examine the stationarity properties of all series. Accordingly, Table 3 reports the results of the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root tests conducted to assess whether the underlying variables are stationary. The unit root tests are initially applied to the variables in levels and subsequently to their first differences. As shown in Table 3, renewable energy adoption and most of the explanatory variables are non-stationary in levels but become stationary after first differencing, indicating that they are integrated of order one, $I(1)$. In contrast, industrial activity is found to be stationary at level, implying an integration order of $I(0)$. The findings confirm a mixture of $I(0)$ and $I(1)$ variables, thereby justifying the use of the NARDL approach, which is appropriate for analysing both short-run and long-run dynamics in the presence of mixed orders of integration, provided that none of the variables is integrated of order two.

Due to the inability of conventional unit root tests such as the Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) tests to capture potential structural changes in the data, this study additionally employs unit root tests

allowing for endogenous structural breaks. The results of these tests are reported in Table 4. Specifically, the ADF test with an endogenously determined structural break of the innovational outlier type, following Vogelsang [48], is applied to all variables. This approach enables the identification of gradual structural shifts that may affect the stationarity properties of the series. The break dates are selected by minimizing the Dickey–Fuller t-statistic. The results indicate that renewable electricity, water abstraction, gross domestic product, and industrial activity are non-stationary in levels but become stationary after first differencing, confirming that they are integrated of order one, $I(1)$. In contrast, agricultural activity is stationary at level, indicating an order of integration $I(0)$. Water abstraction becomes stationary at the first difference with statistical significance at least at the 1%, confirming the robustness of its integration order. Overall, these findings corroborate the results obtained from the conventional unit root tests and further confirm the presence of mixed orders of integration, $I(0)$ and $I(1)$. Consequently, the use of the NARDL model is appropriate, as none of the variables is integrated of order two.

Table 4 Unit root test with structural breaks. Source: Authors' computation using EViews 12

Variable	Break	At level	At 1st difference	Order of integration
logRE	2008	−3.889	−6.385***	$I(1)$
logWA	2002	−3.089	−7.321***	$I(1)$
logGDP	2003	−3.331	−9.274***	$I(1)$
logAGR	1999	−8.343***	−13.907***	$I(0)$
logIND	1997	−3.839	−9.129***	$I(1)$

*** = denotes significance at the 1%

The ADF test is estimated with an endogenous structural break of the innovational outlier type following Vogelsang [48]. The break date is determined by minimizing the Dickey–Fuller test statistic

Cointegration analysis

Before estimating the long-run and short-run dynamics, the NARDL bounds test in Table 5 was applied to verify the existence of a co-integrating relationship among RE, WA, AGR, GDP, and IND. The null hypothesis of no long-run relationship is rejected, as the calculated F-statistic ($F = 6.839$) exceeds the upper critical bound at all conventional significance levels (1%, 5%, and 10%). This confirms that the variables share a long-term equilibrium relationship, justifying the subsequent estimation of long-run coefficients and short-run dynamics using the NARDL model.

Table 5 Results of bounds test for cointegration. Source: Authors' computation using EViews 12

F-statistic	6.839***	
Maximum lag	2	
Lag order	(1, 3, 3, 1, 1, 2)	
K	5	
Critical values		
Significance level	$I(0)$	$I(1)$
1%	3.06	4.00
2.5%	2.70	3.73
5%	2.39	3.38
10%	2.08	3.00

Critical values are based on Pesaran et al. [25], Case II: restricted intercept and no trend

k = number of regressors in the NARDL model (k = 5), significance levels: *** 1%, ** 5%, and * 10%

Table 6 Long-run NARDL estimates. Source: Authors' computation using EViews 12

Variable	Coefficient	Std. Error	p-value
Constant	43.383	44.675	0.352
logWA_POS	−0.867	0.610	0.183
logWA_NEG	0.058	0.555	0.919
logGDP	0.549	0.183	0.012***
logAGR	5.275	1.598	0.007***
logIND	−5.181	2.546	0.067

Significance levels: *** 1%, ** 5%, and * 10%

NARDL estimation results

The NARDL estimation results reveal important insights into how renewable electricity share in Morocco is associated with changes in water abstraction, agricultural activity, economic growth, and industrial development. In the long run, positive changes in water abstraction slightly reduce renewable electricity share (-0.867 , $p = 0.183$), while negative changes slightly increase it (0.058 , $p = 0.919$). As shown in Table 6, neither of these individual effects is statistically significant, suggesting limited long-run responsiveness of renewable electricity share to changes in water abstraction. AGR exhibits a strong and positive effect on renewable electricity share (5.275 , $p = 0.007$), highlighting that agriculture-related dynamics may play an important role in shaping renewable electricity development. In contrast, IND has a negative but non-significant impact (-5.181 , $p = 0.067$), which may reflect the continued reliance of Moroccan industry on conventional energy sources. GDP exhibits a positive and statistically significant effect on renewable electricity share (0.549 , $p = 0.012$), suggesting that economic development may support investment capacity, infrastructure expansion, and technological progress associated with the energy transition in Morocco.

To further verify the robustness of the long-run relationship between water abstraction and renewable electricity share, this study employs the Fully Modified Ordinary Least Squares (FMOLS) estimator. The results presented in Table 7 indicate that water abstraction (WA) does not exert a statistically significant long-run effect on renewable electricity share (REA) in Morocco. Although the estimated coefficient is positive (0.034), the associated probability value ($p = 0.834$) indicates that the effect is statistically insignificant. This finding confirms the long-run NARDL results, which also suggest limited long-run responsiveness of renewable electricity share to changes in water abstraction.

Regarding the control variables, GDP exhibits a positive and highly significant effect on renewable electricity share ($\beta = 1.88E-10$; $p < 0.01$), suggesting that economic growth may support renewable electricity development through greater investment capacity, infrastructure expansion, and technological advancement. Agricultural activity (AGR) also displays a positive and statistically significant relationship with renewable electricity share ($\beta = 1.271$; $p < 0.01$), highlighting the importance of agriculture-related dynamics within Morocco's energy transition process. In contrast, industrial activity (IND) remains statistically insignificant ($p = 0.485$), suggesting that industrial development may still depend largely on conventional energy sources.

The use of FMOLS and DOLS estimators is widely recommended for obtaining consistent long-run estimates in the presence of endogeneity and serial correlation problems. The previous studies have successfully employed these estimators in environmental and energy economics applications to assess the robustness of cointegration relationships [15]. The FMOLS results reinforce the robustness and consistency of the main long-run findings obtained from the NARDL

Table 7 FMOLS long-run estimates. Source: Authors' computation using EViews 12

Variable	Coefficient	Std. Error	t-Statistic	Prob
WA	0.034943	0.165344	0.211334	0.8343
GDP	1.88E−10	2.80E−11	6.722824	0.0000***
AGR	1.271337	0.322647	3.940334	0.0005***
IND	0.301867	0.426530	0.707729	0.4854
C	−27.60779	17.41195	−1.585566	0.1249
Diagnostic statistics				Value
R-squared				0.717031
Adjusted R-squared				0.673498
S.E. of regression				2.674190

Dependent variable: REA. Significance levels: *** 1%, ** 5%, and * 10%

Table 8 DOLS long-run estimates. Source: Authors' computation using EViews 12

Variable	Coefficient	Std. Error	t-Statistic	Prob
WA	0.207308	0.240981	0.860267	0.4065
GDP	3.16E-10	6.33E-11	4.985339	0.0003***
AGR	2.794851	0.981354	2.847953	0.0147**
IND	-1.422447	0.972965	-1.461971	0.1694
C	-17.74324	30.79539	-0.576166	0.5752
Diagnostic statistics				Value
R-squared				0.842908
Adjusted R-squared				0.633452
S.E. of regression				2.799886

Dependent variable: REA. Significance levels: *** 1%, ** 5%, and * 10%

estimation. The results suggest that water abstraction does not constitute a strong long-run determinant of renewable electricity share in Morocco. FMOLS and DOLS are linear cointegration techniques, water abstraction is included in its aggregate form rather than decomposed into positive and negative components. Therefore, these estimators are used to assess the robustness of the overall long-run relationship rather than asymmetric effects. Note that GDP is entered in levels rather than natural logarithms in the FMOLS and DOLS specifications, unlike the NARDL model; consequently, the FMOLS/DOLS GDP coefficient is not directly comparable in magnitude to the NARDL long-run GDP coefficient reported above, although both indicate a positive and statistically significant relationship.

To provide an additional robustness assessment of the long-run relationship, the study also applies the Dynamic Ordinary Least Squares (DOLS) estimator. The results reported in Table 8 remain largely consistent with the NARDL and FMOLS findings.

The coefficient associated with water abstraction (WA) is positive (0.207) but statistically insignificant ($p = 0.406$), indicating that changes in water abstraction do not exert a robust long-run effect on renewable electricity share in Morocco. This consistency across different cointegration estimators strengthens the reliability of the empirical findings and confirms the limited long-run responsiveness of renewable electricity share to changes in water abstraction.

GDP continues to display a positive and statistically significant relationship with renewable electricity share ($\beta = 3.16E-10$; $p < 0.01$), suggesting that economic growth may facilitate renewable electricity expansion through financial resources, infrastructure development, and technological progress. Similarly, agricultural activity (AGR) remains positively and significantly associated

with renewable electricity share ($\beta = 2.794$; $p < 0.05$), confirming the relevance of agricultural dynamics in supporting renewable electricity development. In contrast, industrial activity (IND) remains statistically insignificant ($p = 0.169$), indicating that the industrial sector may still rely predominantly on conventional energy systems during the study period.

The DOLS estimation confirms the robustness and stability of the long-run empirical results obtained from the NARDL and FMOLS approaches.

In the short run, renewable electricity share reacts more strongly to immediate changes in water abstraction. Positive short-term changes increase renewable electricity share slightly (1.035, $p = 0.015$), while negative changes reduce it substantially (-5.275 , $p = 0.000$), indicating a clear asymmetric response in the short run. As shown in Table 9, this asymmetry suggests that negative changes in water abstraction are more strongly associated with short-run variations in renewable electricity share than positive changes. Such reductions in water abstraction may reflect periods of water stress, policy restrictions, or adjustments in water demand. Agricultural activity remains a positive and highly significant factor associated with renewable electricity share ($\Delta AGR = 2.132$, $p = 0.000$), confirming the importance of sectoral dynamics in Morocco's energy transition. The error correction term ($ECM(-1) = -0.570$, $p < 0.01$) indicates a relatively rapid adjustment towards long-run equilibrium, with approximately 57% of short-run deviations corrected in each period. This finding is supported by the diagnostic statistics, as the model suggests an adequate statistical fit ($R^2 = 0.923$; adjusted $R^2 = 0.878$) and no evidence of serial correlation in the residuals (Durbin-Watson = 2.241), indicating reasonably stable residual behaviour.

Table 9 Short-run NARDL and ECM. Source: Authors' computation using EViews 12

Variable	Coefficient	Std. Error	p-value
$\Delta \log WA_POS$	1.035	0.358	0.015***
$\Delta \log WA_POS(-1)$	-0.531	0.438	0.251
$\Delta \log WA_POS(-2)$	-1.501	0.384	0.002***
$\Delta \log WA_NEG$	-5.275	0.721	0.000***
$\Delta \log WA_NEG(-1)$	1.817	0.866	0.060
$\Delta \log WA_NEG(-2)$	3.022	0.675	0.001***
$\Delta \log GDP$	0.578	0.803	0.487
$\Delta \log AGR$	2.132	0.204	0.000***
$\Delta \log IND$	-0.879	0.292	0.012***
$\Delta \log IND(-1)$	1.681	0.271	0.000***
ECM(-1)	-0.570	0.066	0.000***
Diagnostic statistics			
R-squared (R^2)			0.923
Adjusted R^2			0.878
Durbin-Watson			2.241

Significance levels: *** 1%, ** 5%, and * 10%

Table 10 Robustness check of the lag structure. Source: Authors' computation using EViews 12

	Original (1,3,3,1,1,2)	Parsimonious (1,1,1,1,1,1)
Observations	28	30
Estimated parameters	23	18
Degrees of freedom	5	12
Adjusted R^2	0.932	0.415
ECM coefficient (p-value)	-0.890 (0.009)	-0.873 (0.006)
Short-run asymmetry test (p-value)	0.137	0.347

Robustness to lag specification

To assess the sensitivity of the results to the richness of the lag structure, the NARDL model was re-estimated using a more parsimonious specification, replacing the AIC-selected lag order (1, 3, 3, 1, 1, 2) with a uniform lag

order of one for all variables. Table 10 compares the two specifications.

As a robustness check, the NARDL specification was re-estimated using a more parsimonious lag structure (lag order 1 for all variables) instead of the original AIC-selected specification (1, 3, 3, 1, 1, 2), which uses 23 parameters for only 28 usable observations (5 degrees of freedom). The error-correction term remains negative and statistically significant across specifications, confirming that the equilibrium-correction mechanism is not an artefact of the richer lag structure. The direction of the short-run asymmetry (a positive effect of increases in water abstraction and a negative effect of decreases) is preserved under the parsimonious specification, although its statistical significance is more limited, reflecting the reduced degrees of freedom available with 32 annual observations. These results support the qualitative robustness of the study's main conclusions while underscoring the need for

Table 11 Wald test for asymmetry. Source: Authors' computation using EViews 12

Term	Null hypothesis	Normalized restriction (= 0)	Std. Error	F-statistic	Probability
Long run	$C(2) = C(6)$	6.309	1.562	16.30	0.00***
Short run	$C(3) = C(7)$,	$C(3) - C(7) = -0.465$	1.026	5.33	0.01*** (F)
	$C(4) = C(8)$,	$C(4) - C(8) = 3.214$	1.184		0.00*** (χ^2)
	$C(5) = C(9)$	$C(5) - C(9) = -8.094$	2.053		

Significance levels: *** 1%, ** 5%, * 10% indicate asymmetry

C(2)–C(6): long-run coefficients of WA_POS and WA_NEG

C(3), C(4), C(5) – C(7), C(8), C(9): short-run coefficients of lagged WA_POS and WA_NEG differences

Null hypothesis assumes symmetry: Positive and negative water changes have the same effect

caution when interpreting the precise magnitude of the short-run asymmetric effects.

Asymmetry analysis

The results suggest that renewable electricity share in Morocco exhibits limited long-run responsiveness to changes in water abstraction, whereas short-run dynamics appear more asymmetric. In particular, negative changes in water abstraction are associated with stronger short-run responses than positive changes. Agricultural activity remains an important factor associated with renewable electricity share, while the industrial sector appears to rely more heavily on conventional energy sources, indicating potential opportunities for future renewable energy integration. To determine whether the long-run and short-run responses are symmetric or asymmetric, a Wald test is conducted to examine whether positive and negative water shocks exert similar effects on renewable electricity share in Morocco. The results reported in Table 11 reject the null hypothesis of symmetry ($F = 16.31$, $p = 0.002$; $\chi^2 = 16.00$, $p = 0.0011$), indicating that positive and negative changes in water

abstraction do not affect renewable electricity share in the same manner. Although the estimated long-run coefficients associated with positive and negative changes in water abstraction are individually statistically insignificant, the Wald test suggests that the overall adjustment process may still exhibit asymmetric properties when positive and negative shocks are considered jointly. This finding indicates that the detected asymmetry is better interpreted as a nonlinear adjustment mechanism rather than as evidence of strong individual long-run effects. In the short run, negative changes in water abstraction are more strongly associated with renewable electricity share than positive changes, highlighting the presence of asymmetric adjustment dynamics. However, these findings should be interpreted cautiously, as water abstraction may capture several underlying economic, institutional, and environmental processes. In the long run, the influence of water abstraction appears more limited, suggesting that structural factors such as energy policies, investment capacity, and infrastructure development play a more central role in shaping Morocco's renewable electricity transition. Overall, the Wald test supports the

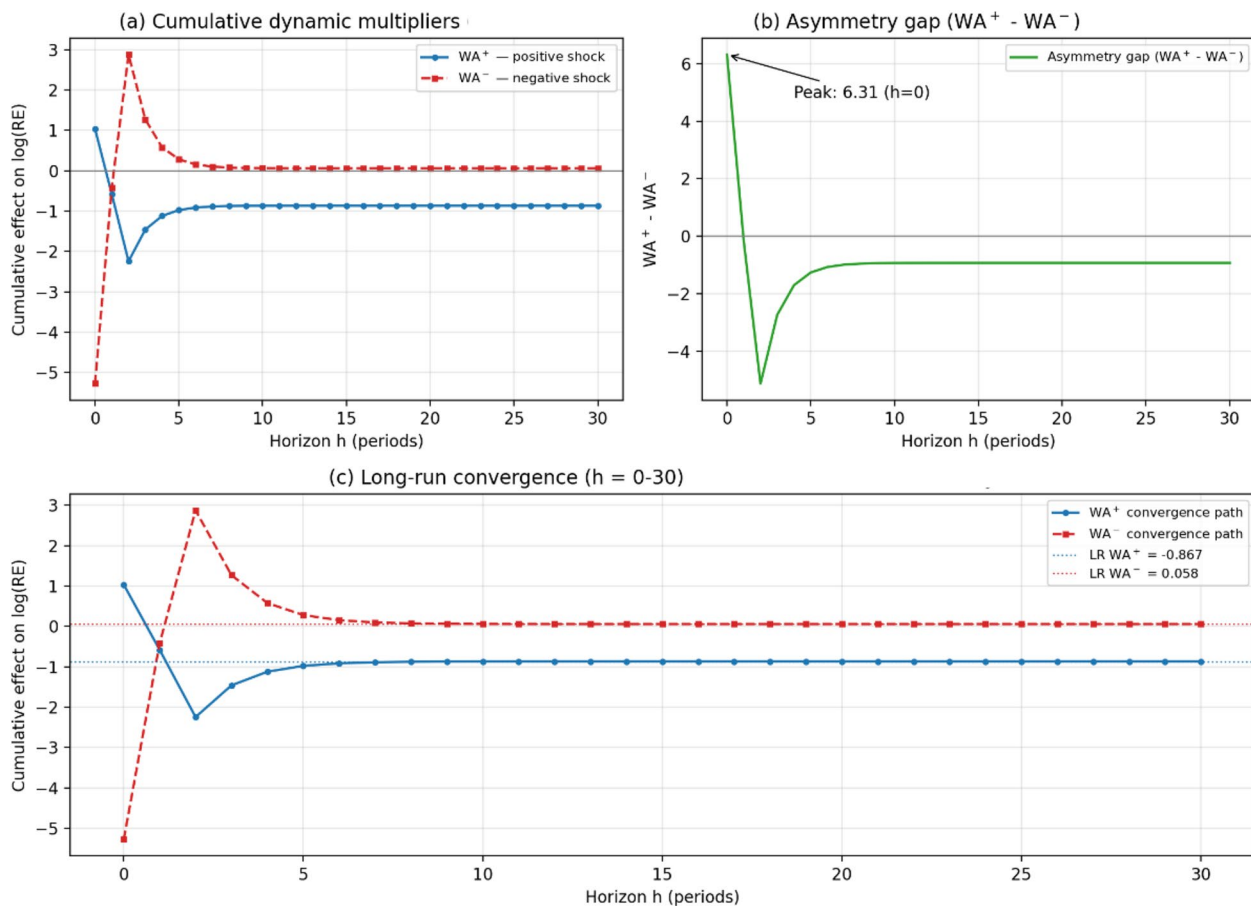


Fig. 5 Dynamic multipliers. Source: Authors' computation using EViews 12

presence of asymmetric adjustment dynamics between water abstraction and renewable electricity share, particularly in the short run, which further justifies the use of the NARDL framework to capture these nonlinear relationships.

Dynamic multiplier analysis

To further explore the asymmetric relationship between water abstraction and renewable electricity share, Fig. 5 presents the cumulative dynamic multipliers associated with positive and negative changes in water abstraction. These multipliers are computed by tracing the response of $\log(\text{RE})$ to a one time, permanent unit increase in the level of WA^+ (respectively, WA^-), propagated forward through the estimated short-run NARDL equation reported in Table 9, including the error-correction feedback term. By construction, each multiplier path converges, as the horizon increases, to the corresponding long-run coefficient reported in Table 6 ($\beta_{\text{xa}}^+ = -0.867$ and $\beta_{\text{xa}}^- = 0.058$).

Panel (a) displays the cumulative responses of renewable electricity share to positive (WA^+) and negative (WA^-) shocks in water abstraction. The figure reveals markedly different and distinctly non-monotonic, adjustment paths during the periods following the shock. A positive shock initially generates a favourable response, with the cumulative multiplier reaching approximately 1.04 at the impact period ($h = 0$). This effect then declines sharply, turning negative by $h = 1$ and reaching a trough of approximately -2.24 at $h = 2$, before gradually recovering and converging towards its long-run value of -0.87 . In contrast, a negative shock produces a much larger initial response, with the cumulative effect reaching approximately -5.28 at the impact period. This effect then reverses sharply, overshooting into positive territory with a peak of approximately $+2.88$ at $h = 2$, before gradually declining and converging towards its long-run value of $+0.06$. This result is consistent with the short-run NARDL estimates reported in Table 9, where negative changes in water abstraction exert a considerably stronger contemporaneous effect than positive changes, while the lagged short-run terms (significant up to the second lag for both WA^+ and WA^-) generate the pronounced overshoot observed around $h = 2$ before the error-correction mechanism brings the system back towards equilibrium.

The evolution of the asymmetry itself is illustrated in Panel (b), which reports the gap between the cumulative responses to positive and negative shocks. The asymmetry gap reaches its maximum value of approximately 6.31 at the impact period ($h = 0$), reflecting the much larger contemporaneous response to a negative shock relative to

a positive one. Unlike a gradual narrowing, however, the gap changes sign almost immediately, turning negative from $h = 1$ onward and reaching a pronounced trough of approximately -5.12 at $h = 2$, the point at which the overshoot in the WA^- response is largest relative to the trough in the WA^+ response. Thereafter, the gap stabilizes rapidly, settling near its long-run value of approximately -0.93 (the difference between the two long-run coefficients, $\beta_{\text{xa}}^+ - \beta_{\text{xa}}^- = -0.867 - 0.058$) from around $h = 8$ onward. This pattern indicates that the asymmetry between positive and negative shocks is not a slowly fading phenomenon but one that reverses direction quickly, with the system briefly overshooting before settling into a persistent, smaller long-run asymmetry.

This finding provides precisely the type of information that cannot be obtained from coefficient estimates alone. Although the Wald test confirms the existence of asymmetry, the dynamic multipliers reveal how this asymmetry evolves through time not as a monotonic decay, but as a rapid reversal and partial overshoot before stabilization. Panel (c) presents the convergence of the cumulative multipliers towards their long-run equilibrium values. The adjustment process is relatively rapid: After the initial oscillation described above, both trajectories settle close to their respective long-run levels by approximately $h = 10$, and are indistinguishable from those levels by $h = 15$. The cumulative response associated with positive shocks converges to -0.867 , exactly matching the long-run NARDL coefficient reported in Table 6, and the response associated with negative shocks converges to 0.058 , likewise matching Table 6 exactly. This exact correspondence is a direct mathematical consequence of the error-correction representation of the NARDL model [7] and confirms the internal consistency of the short-run and long-run estimates. The relatively fast convergence observed in the figure is consistent with the estimated error-correction coefficient reported in Table 9 ($\text{ECM} = -0.570$, $p < 0.01$), which implies that approximately 57% of any deviation from the long-run equilibrium is corrected within one period; combined with the significant second-lag short-run coefficients, this feedback mechanism produces the overshoot-and-correct pattern visible in Panel (a) rather than a smooth, monotonic return to equilibrium.

An additional insight emerging from Fig. 5 concerns the persistence of the effects. While the largest positive-negative divergence occurs very early (at or immediately after impact), neither shock disappears immediately, and the system passes through a corrective overshoot before settling. This pattern suggests that water-related disturbances have consequences that extend for several periods beyond the initial shock, even though the direction and magnitude of these consequences change substantially

over the adjustment path. Taken together, the dynamic multiplier analysis strengthens the evidence obtained from the NARDL estimation and Wald asymmetry tests. The results show that renewable electricity share responds very differently, and non-monotonically, to positive and negative changes in water abstraction, that the sharpest divergence occurs within the first two periods, and that the system converges rapidly and exactly, by construction to the long-run equilibrium values reported in Table 6. These findings highlight the importance of accounting not only for the existence of asymmetry but also for its short-run, oscillatory temporal evolution, as the adjustment process itself provides valuable information regarding the resilience of the water–energy nexus in Morocco.

The interpretation of the estimated asymmetries requires caution. Positive and negative changes in water abstraction represent variations in water withdrawal patterns and may reflect a combination of economic, environmental, and institutional factors. Consequently, the estimated asymmetries including the short-lived overshoot documented above should be interpreted as differences in adjustment dynamics rather than as direct responses to water scarcity or water abundance.

Diagnostic and stability tests

Given the importance of these findings, a series of diagnostic and stability tests were conducted to evaluate the reliability and adequacy of the estimated NARDL model. The results reported in Table 12 indicate that the residuals are normally distributed and free from serial correlation, suggesting that the dynamic specification of the model is appropriate. The diagnostic tests also confirm the absence of heteroskedasticity and ARCH effects, implying that the variance of the error term remains stable over time and that the estimated coefficients are statistically reliable. In addition, the Ramsey RESET test indicates that the model is correctly specified and does not suffer from functional form misspecification. Finally, the CUSUM and CUSUMSQ tests presented in Figs. 6 and 7 show that the estimated coefficients remain stable

Table 12 Diagnostic tests of the NARDL model. Source: Authors' computation using EViews 12

Diagnostic test	Statistic	p-value
Residual normality (Jarque–Bera)	2.189	0.3345
Serial correlation (Breusch–Godfrey LM)	0.663	0.5389
Heteroskedasticity (Breusch–Pagan–Godfrey)	0.678	0.7669
ARCH effect	1.326	0.2603
Model specification (Ramsey RESET)	0.019	0.8944

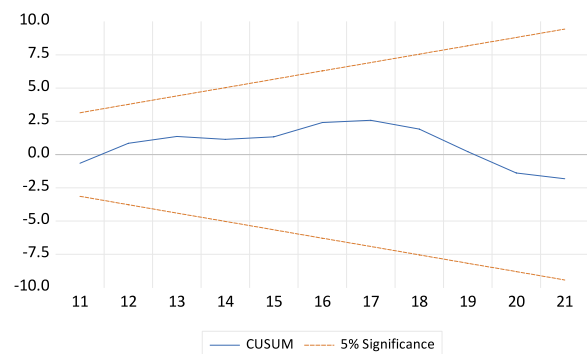


Fig. 6 Cumulative sum (CUSUM) test. Source: Authors' computation using EViews 12

throughout the sample period, suggesting the absence of significant structural instability in the estimated model.

Asymmetric causality test

Moreover, Table 13 presents the results of the asymmetric Granger causality test, which helps explain the dynamic relationship between water abstraction and renewable electricity share in Morocco. This test distinguishes between positive and negative changes in water abstraction, allowing for the analysis of potential asymmetric effects.

The results indicate the presence of a one-way causal relationship running from negative changes in water abstraction to renewable electricity share. This finding suggests that reductions in water abstraction are followed by changes in renewable electricity share. However, the reverse causal relationship is not supported by the results, indicating that renewable electricity share does not appear to influence water abstraction in the short run.

These findings may suggest that periods associated with reduced water abstraction, which can reflect water

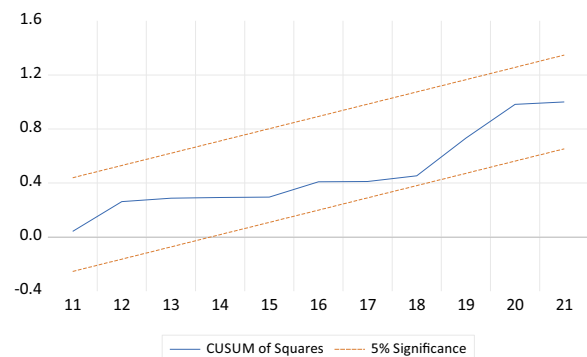


Fig. 7 Cumulative sum of squares (CUSUMSQ) test. Source: Authors' computation using EViews 12

Table 13 Asymmetric causality test. Source: Authors' computation using EVIEWS 12

Null hypothesis	F-statistic	p-value	Result	Direction of causality
WA_POS does not Granger Cause REA	1.770	0.192	Do not Reject H0	–
REA does not Granger Cause WA_POS	0.330	0.722	Do not Reject H0	–
WA_NEG does not Granger Cause REA	4.969	0.016	Reject H0	WA_NEG → REA
REA does not Granger Cause WA_NEG	1.345	0.280	Do not Reject H0	–
GDP does not Granger Cause REA	8.597	0.001	Reject H0	GDP → REA
REA does not Granger Cause GDP	2.429	0.109	Do not Reject H0	–
AGR does not Granger Cause REA	2.410	0.110	Do not Reject H0	–
REA does not Granger Cause AGR	1.509	0.241	Do not Reject H0	–
IND does not Granger Cause REA	2.269	0.124	Do not Reject H0	–
REA does not Granger Cause IND	1.931	0.166	Do not Reject H0	–
WA_NEG does not Granger Cause WA_POS	0.331	0.721	Do not Reject H0	–
WA_POS does not Granger Cause WA_NEG	8.485	0.002	Reject H0	WA_POS → WA_NEG
GDP does not Granger Cause WA_POS	0.177	0.839	Do not Reject H0	–
WA_POS does not Granger Cause GDP	4.671	0.019	Reject H0	WA_POS → GDP
AGR does not Granger Cause WA_POS	4.331	0.025	Reject H0	AGR → WA_POS
WA_POS does not Granger Cause AGR	2.536	0.100	Do not Reject H0	–
IND does not Granger Cause WA_POS	0.358	0.703	Do not Reject H0	–
WA_POS does not Granger Cause IND	0.713	0.500	Do not Reject H0	–
GDP does not Granger Cause WA_NEG	2.290	0.123	Do not Reject H0	–
WA_NEG does not Granger Cause GDP	2.374	0.115	Do not Reject H0	–
AGR does not Granger Cause WA_NEG	2.992	0.069	Do not Reject H0	–
WA_NEG does not Granger Cause AGR	0.257	0.776	Do not Reject H0	–
IND does not Granger Cause WA_NEG	1.610	0.221	Do not Reject H0	–
WA_NEG does not Granger Cause IND	1.420	0.261	Do not Reject H0	–
AGR does not Granger Cause GDP	3.606	0.042	Reject H0	AGR → GDP
GDP does not Granger Cause AGR	2.623	0.093	Do not Reject H0	–
IND does not Granger Cause GDP	3.007	0.068	Do not Reject H0	–
GDP does not Granger Cause IND	0.891	0.423	Do not Reject H0	–
IND does not Granger Cause AGR	1.026	0.373	Do not Reject H0	–
AGR does not Granger Cause IND	1.472	0.249	Do not Reject H0	–

stress conditions, drought episodes, policy restrictions, or adjustments in water demand, are linked to subsequent changes in renewable electricity share. In contrast, positive changes in water abstraction do not exhibit a significant causal effect on renewable electricity share. This may indicate that energy-related adjustments respond more strongly to constrained water conditions than to periods characterized by relatively higher water use (Fig. 8).

The results also reveal a one-way causality running from GDP to renewable electricity share. This suggests that economic growth may contribute to renewable electricity development through improved investment capacity, infrastructure expansion, and financial resources. However, no reverse causality from renewable electricity share to GDP is detected during the study period. For agricultural and industrial activities, no statistically significant causal relationship with renewable electricity

share is identified. Although these sectors are strongly connected to water and energy use, short-run fluctuations in agricultural and industrial activity do not appear to directly influence renewable electricity share. This may reflect the continued dependence of these sectors on conventional energy sources and the relatively gradual integration of renewable technologies. The asymmetric Granger causality results suggest that renewable electricity share in Morocco responds more strongly to constrained water conditions than to positive changes in water abstraction. These findings highlight the importance of considering water resource conditions within energy planning strategies. Policies aimed at improving water efficiency and reducing pressure on water resources may, therefore, contribute to supporting a more sustainable energy transition in Morocco.

The results provide partial support for the study's hypotheses. The long-run estimates indicate that changes

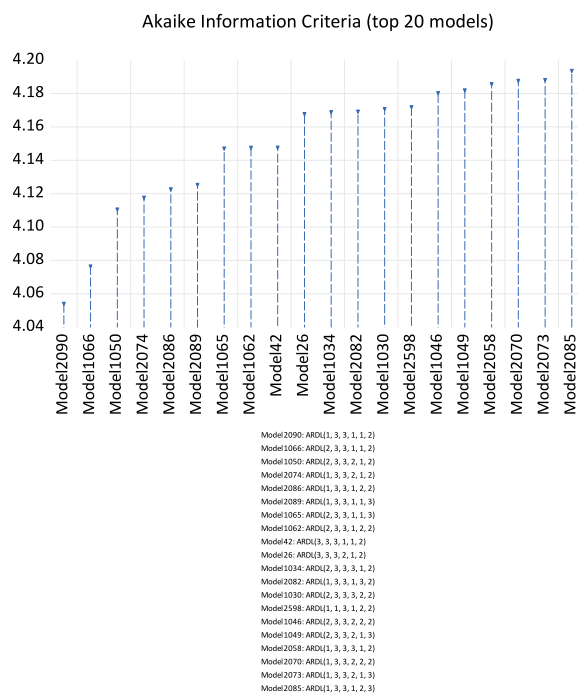


Fig. 8 Model selection. Source: Authors' computation using EViews 12

in water abstraction do not exert a statistically significant effect on renewable electricity share, suggesting limited long-run responsiveness. By contrast, the short-run results reveal asymmetric dynamics, as renewable electricity share responds differently to positive and negative changes in water abstraction. In particular, negative changes in water abstraction are associated with stronger short-run responses than positive changes.

These findings are further supported by the asymmetric Granger causality analysis, which identifies a unidirectional causal relationship running from negative changes in water abstraction to renewable electricity share. The evidence, therefore, points to the existence of nonlinear short-run adjustment dynamics linking water abstraction and renewable electricity share. However, given that the individual long-run coefficients for positive and negative water abstraction are not statistically significant, the results suggest that these interactions are primarily short-run in nature rather than indicative of a persistent long-run relationship. It should be noted that this insignificance of the individual long-run coefficients is distinct from the Wald test of long-run symmetry, which tests whether the two coefficients differ from each other rather than whether either is individually different from zero; the two findings are not contradictory but capture different aspects of the long-run relationship.

Discussion of findings

The empirical evidence suggests that water abstraction does not exert a statistically significant long-run effect on renewable electricity share in Morocco. This conclusion is consistently supported by the NARDL, FMOLS, and DOLS estimations, thereby strengthening the robustness of the findings. Although this result may appear unexpected within the broader water–energy nexus literature, several structural and contextual factors may explain the absence of a strong long-run relationship. A first explanation relates to the technological composition of Morocco's renewable energy portfolio. During the past two decades, the expansion of renewable electricity generation has been driven predominantly by solar photovoltaic and wind technologies, whose operational water requirements remain substantially lower than those associated with hydropower generation, bioenergy production, or concentrated solar power facilities. While certain renewable technologies remain highly dependent on water availability, Morocco's renewable energy strategy has progressively diversified towards technologies characterized by relatively limited freshwater dependence. Consequently, aggregate changes in water abstraction may not constitute a sufficiently strong determinant capable of influencing long-run renewable electricity trajectories. A second explanation concerns the institutional dimension of Morocco's energy transition. Renewable electricity deployment in Morocco has largely been shaped by proactive public policies, long-term investment programmes, and strategic national commitments rather than by fluctuations in resource availability alone. Large-scale initiatives implemented under the national energy strategy, combined with substantial investments in solar and wind infrastructure, have contributed to decoupling renewable electricity expansion from short-term variations in water conditions. This interpretation is consistent with the significant positive effect observed for GDP, suggesting that financial capacity, infrastructure development, and technological investments remain among the principal drivers of renewable electricity growth.

The insignificant long-run relationship may also reflect the aggregate nature of the water abstraction indicator used in this study. Water abstraction encompasses withdrawals from agricultural, industrial, domestic, and environmental uses and, therefore, captures multiple underlying processes simultaneously. As a consequence, variations in aggregate water abstraction do not necessarily translate directly into changes in renewable electricity generation. In the Moroccan context, agricultural activities account for the dominant share of water consumption, whereas the renewable electricity sector represents a comparatively smaller component of total water demand. This sectoral imbalance may partly explain why

changes in aggregate water withdrawals fail to produce statistically significant long-run effects on renewable electricity share.

Furthermore, Morocco has increasingly adopted adaptation measures aimed at reducing the vulnerability of its energy transition to water constraints. The expansion of desalination capacities, improvements in irrigation efficiency, inter-basin water transfer projects, and the development of integrated water management policies may have attenuated the long-run dependence between water availability and renewable electricity production. Such adaptation mechanisms may allow the renewable electricity sector to maintain its growth trajectory despite increasing pressure on freshwater resources.

Nevertheless, the absence of statistically significant long-run effects should not be interpreted as evidence that water conditions are irrelevant for renewable electricity development. On the contrary, the short-run results reveal the existence of pronounced asymmetric adjustment dynamics. Positive changes in water abstraction are associated with modest increases in renewable electricity share, whereas negative changes generate substantially larger adverse effects. This finding suggests that renewable electricity systems are more sensitive to constrained water conditions than to periods characterized by relatively abundant water availability. This asymmetry appears particularly relevant in Morocco, where increasing climatic variability, declining precipitation levels, and recurrent drought episodes have become structural features of the national environmental context. Periods associated with reductions in water abstraction may reflect episodes of water stress, drought-induced restrictions, or reallocation mechanisms across competing sectors. Under such conditions, energy-related decisions may experience temporary disruptions, leading to stronger short-run adjustments in renewable electricity dynamics.

The dynamic multiplier analysis reinforces this interpretation while revealing a more active adjustment mechanism. The divergence between positive and negative shocks is greatest at impact and, rather than diminishing gradually, reverses within a single period, generating a pronounced but short-lived overshoot. Specifically, the response to positive water withdrawal shocks (WA^+) reaches a trough of approximately -2.24 , whereas the response to negative shocks (WA^-) peaks at about $+2.88$ around $h = 2$, before both trajectories converge towards their respective long-run equilibria. This pattern suggests that the Moroccan renewable electricity sector does not respond to water-related disturbances through a simple monotonic adjustment, but rather through a rapid oscillatory correction process. Such dynamics are consistent with the relatively strong error-correction mechanism

($ECM = -0.570$) and the statistically significant second-lag short-run effects identified for both WA^+ and WA^- . Substantively, these findings reinforce the paper's central argument that the asymmetry between positive and negative water withdrawal shocks is confined to the short run and disappears as the system converges to its long-run equilibrium. Consequently, the limited long-run responsiveness reported in Table 6 and corroborated by the FMOLS and DOLS estimations remains valid. Overall, the rapid yet convergent adjustment dynamics indicate that, although Morocco's renewable electricity sector is temporarily sensitive to fluctuations in water withdrawal patterns, it is not structurally dependent on them in the long run.

The asymmetric causality results provide additional support for this interpretation. The existence of a unidirectional causal relationship running from negative changes in water abstraction towards renewable electricity share indicates that renewable electricity systems react more strongly to adverse water conditions than to improvements in water availability. This result reinforces the argument that water-related constraints operate primarily through short-run adjustment mechanisms rather than through persistent long-run structural effects.

These findings, therefore, contribute to the growing literature on the water–energy nexus by suggesting that the relationship between water resources and renewable electricity generation may be highly context-dependent. While the previous studies have reported significant interdependencies between water availability and renewable energy development, particularly in hydropower-dominated systems and water-intensive renewable technologies, the Moroccan case highlights a different pattern. Renewable electricity expansion appears to be driven primarily by economic development, institutional support, technological progress, and strategic investments, whereas water conditions mainly influence short-term adjustment dynamics.

From a policy perspective, these results imply that Morocco's renewable energy transition has already developed a certain degree of resilience to long-run fluctuations in water availability. However, the existence of significant short-run asymmetries suggests that water resource management should remain an integral component of energy planning strategies. Strengthening integrated water–energy governance frameworks, promoting low-water renewable technologies, expanding renewable-powered desalination projects, and incorporating drought scenarios into energy planning processes may contribute to enhancing the resilience of Morocco's low-carbon transition under increasing climatic uncertainty.

Conclusion and policy propositions

This study investigated the relationship between water abstraction and renewable electricity share in Morocco over the period 1990–2021 using a Nonlinear Autoregressive Distributed Lag (NARDL) framework combined with an asymmetric Granger causality approach. The analysis aimed to determine whether variations in water abstraction influence renewable electricity generation and whether positive and negative changes in water abstraction exert asymmetric effects.

The empirical findings reveal the existence of a long-run equilibrium relationship among renewable electricity share, water abstraction, economic growth, agricultural activity, and industrial development. However, the estimated long-run coefficients indicate that neither positive nor negative changes in water abstraction exert statistically significant effects on renewable electricity share. These findings are consistently supported by the FMOLS and DOLS estimations, reinforcing the robustness of the results and suggesting that renewable electricity development in Morocco exhibits limited long-run responsiveness to aggregate variations in water abstraction. The absence of a significant long-run relationship should not be interpreted as evidence that water resources are irrelevant for renewable electricity development. Rather, it suggests that Morocco's renewable electricity transition has been primarily driven by structural factors, including sustained public investment, long-term energy policies, technological progress, and the increasing deployment of solar photovoltaic and wind technologies, which generally exhibit relatively low operational water requirements. Moreover, water abstraction is measured as an aggregate indicator encompassing agricultural, industrial, domestic, and environmental uses, which may partly explain the limited long-run association observed between water withdrawals and renewable electricity generation. By contrast, the short-run results reveal pronounced asymmetric adjustment dynamics. Positive changes in water abstraction exert relatively modest effects on renewable electricity share, whereas negative changes generate substantially stronger responses. The Wald tests confirm the presence of significant asymmetries, while the dynamic multiplier analysis demonstrates that these differences are most pronounced during the initial adjustment periods and gradually disappear as the system converges towards equilibrium. Furthermore, the asymmetric causality analysis identifies a unidirectional causal relationship running from negative changes in water abstraction to renewable electricity share, suggesting that renewable electricity systems respond more strongly to constrained water conditions than to periods characterized by higher water availability.

These findings are particularly relevant in the Moroccan context, where increasing climatic variability, declining precipitation patterns, recurrent drought episodes,

and growing competition among economic sectors have intensified pressure on freshwater resources. The results indicate that water-related constraints operate primarily through short-run adjustment mechanisms rather than through persistent long-run effects. Consequently, renewable electricity expansion in Morocco appears relatively resilient to long-run variations in aggregate water abstraction, while remaining vulnerable to temporary water-related disturbances.

Regarding the control variables, economic growth exerts a positive and significant influence on renewable electricity share, highlighting the role of investment capacity, infrastructure expansion, and technological development in supporting the energy transition process. Agricultural activity also exhibits a strong positive association with renewable electricity generation, suggesting that transformations occurring within water-intensive sectors may contribute indirectly to renewable electricity development. Conversely, industrial activity does not display a statistically significant long-run effect, reflecting the continued dependence of Moroccan industries on conventional energy sources and indicating substantial opportunities for accelerating renewable energy integration within industrial production systems. From a policy perspective, several implications emerge from these findings. First, renewable electricity planning should be more closely integrated with water resource management strategies. Given the sensitivity of renewable electricity share to constrained water conditions in the short run, greater coordination between water authorities and energy planners is essential to improve resource allocation efficiency and strengthen climate resilience. Second, policymakers should continue promoting renewable energy technologies characterized by relatively low water requirements. Expanding solar photovoltaic and wind energy capacities may reduce pressure on freshwater resources while simultaneously supporting national decarbonization objectives. Third, improving water-use efficiency should be considered an important complementary instrument for facilitating sustainable energy transitions. Investments in efficient irrigation systems, reduction of distribution losses, wastewater reuse, and water-saving technologies may contribute to reducing resource pressures and enhancing long-term energy security. Fourth, renewable-powered desalination projects should be further encouraged to address growing water stress challenges. Given Morocco's increasing reliance on desalination infrastructure, coupling desalination facilities with renewable electricity generation could simultaneously strengthen water stress and support low-carbon development objectives. Fifth, the industrial sector should become a strategic priority within future renewable energy policies. The insignificant relationship observed between industrial activity and renewable electricity share suggests that renewable energy penetration within industrial processes remains limited. Expanding

energy efficiency programmes, encouraging cleaner production technologies, and providing incentives for renewable energy adoption within industrial activities could significantly contribute to accelerating Morocco's energy transition.

This study contributes to the growing literature on the water–energy nexus by demonstrating that the relationship between water abstraction and renewable electricity generation cannot be adequately captured through conventional linear approaches. While aggregate water abstraction does not appear to constitute a major long-run determinant of renewable electricity development in Morocco, the presence of significant short-run asymmetries highlights the importance of accounting for nonlinear adjustment mechanisms when analysing interactions between water resources and energy systems.

Despite these contributions, several limitations should be acknowledged. First, the empirical analysis is based on a relatively small sample consisting of thirty-two annual observations. Although the NARDL framework performs relatively well in small samples, coefficient estimates should, nevertheless, be interpreted with caution. Second, renewable electricity share represents only one dimension of renewable energy development and may not fully capture broader transformations occurring within the energy sector. Third, water abstraction is employed as a proxy for water resource conditions and does not directly measure water scarcity, hydrological stress, or climate vulnerability.

Future research could extend the present analysis by incorporating additional variables related to governance quality, technological innovation, climate risks, institutional effectiveness, and urbanization processes. Further investigations may also consider sector-specific indicators of water use, alternative measures of water stress, and advanced nonlinear modelling techniques to improve understanding of the complex interactions linking water resources and renewable electricity development under conditions of increasing climatic uncertainty.

Abbreviations

WA	Water abstraction
RE	Renewable electricity share
WA_POSS	Water positive
WA_NEG	Water negative
GDP	Gross domestic product
AGR	Agriculture value added to GDP
IND	Industrial value added to GDP
NARDL	Nonlinear autoregressive distributed lag
LR	Long run
SR	Short run
ECM	Error correction model
ADF	Augmented Dickey–Fuller
PP	Phillips–Perron

Author contributions

MN helped in conceptualization, methodology, formal analysis, writing—original draft, and data curation; AA worked in supervision, resources, project administration, writing—review, and validation. All authors have read and agreed to the published version of the manuscript.

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Data availability

The data were collected from the World Development Indicators (2023) [<https://databank.worldbank.org/source/world-development-indicators>].

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

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